
Stefan Henkler

E-Mail: stefan.henkler@hshl.de

► Seminar

- Based on a given topic
- Paper (5 pages) plus seminar presentation
 - IEEE latex template
(<https://www.ieee.org/conferences/publishing/templates.html>)
 - Presentation 10 Min + 5 Min discussion
 - Paper written in scientific way
 - Prerequisite: passed scientific work module
 - We will not repeat details on this.
 - Plagiarism: illegal -> failed, triggers internal process, wc ex-matriculated!
 - I.e. every sentence of the paper is of intellectual origin of the author and not a form of copy (direct or intellectual) from any other author (no matter from which resource).

► Use citavi or Zotero for managing your references

- Write for each related reference a short note of how it is related to your seminar topic – the note is part of your bibtex management tool and/or an extra bibtex entry.

► Use github!

- Rest of semester: preparation of paper/topic and presentation plus lab work
 - Details follows

► Possible topics

Earliest Deadline First
Rate Monotonic Scheduling
Polling Server (incl. Introduction to fixed priority servers)
Deferrable Server
Priority Exchange
Sporadic Server
Slack Stealing
Dynamic Priority Exchange Server
Dynamic Sporadic Server
Total Bandwidth Server
Earliest Deadline Late Server
Improved Priority Exchange Server
Constant Bandwidth Server
Non-Preemptive Scheduling
Handling Aperiodic Overloads
Handling Overruns

- Familiarize yourself with the topic based on the main reference.
- Check additional references
 - Books, paper, ... (see hshl bib, oreilly, ieee, acm, ...)
- The elaboration consists of the following parts
 - Necessary mathematical basics
 - Formal (mathematical) description of the topic
 - Formal description of the algorithm
 - Runtime / overhead of the algorithm
 - Implementation of main feature in UPPAAL (if not possible in C/C++)
 - Realization of an application example
 - If possible, by using an appropriate framework or by implementing (parts of the algorithm) in UPPAAL (C/C++)